

# The Effects of Heavy Resisted Sled Pulling on Sprint Mechanics and Spatiotemporal Parameters

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## Abstract

Stavridis, I, Ekizos, A, Zisi, M, Agilara, G-O, Tsolakis, C, Terzis, G, and Paradisis, G. The effects of heavy resisted sled pulling on sprint mechanics and spatiotemporal parameters. *J Strength Cond Res* 37(12): 2346–2353, 2023—This study examines the effects of 2 resisted sled sprinting (RSS) training programs: with a load corresponding to the running velocity associated with the apex of the individual velocity-power relationship ( $50\%v_{dec}$ ), with a load equal to 10% of body mass (10% BM), and of an unresisted sprint training (URS). We measured the 30-m sprint performance in intervals of 5 m examining sprint acceleration, mechanical properties (theoretical maximal horizontal power [ $P_{max}$ ], force [ $F_0$ ], velocity [ $v_0$ ], slope of the force-velocity relationship [ $S_{Fv}$ ], maximal ratio of horizontal-to-resultant force [ $RF_{max}$ ], rate of decrease in RF [ $D_{rf}$ ]), and spatiotemporal parameters (step frequency [SF], step length [SL], flight time [FT], and contact time [CT]). Twenty-seven sprinters were randomly assigned into the  $50\%v_{dec}$ , 10% BM, and URS groups, performing 12 sessions over 6 consecutive weeks (2 sets of 5 sprints per session). The  $50\%v_{dec}$  group significantly improved ( $p < 0.05$ ) their performance in all 30-m intervals. Posttraining, the  $50\%v_{dec}$  group showed significantly increased  $P_{max}$ ,  $F_0$ , and  $RF_{max}$  (mean differences:  $1.46 \pm 1.70 \text{ W}\cdot\text{kg}^{-1}$ ,  $0.51 \pm 0.68 \text{ N}\cdot\text{kg}^{-1}$ , and  $0.17 \pm 0.18\%$ , respectively), compared with pretraining. The  $50\%v_{dec}$  group achieved higher SF, whereas FT decreased postintervention. No significant changes ( $p > 0.05$ ) were found in the performance and mechanical and spatiotemporal variables in the other groups. In conclusion, RSS training with a load of  $50\%v_{dec}$  provides an effective loading stimulus to induce adaptations that improve sprint acceleration performance. The improvements are explained by greater amounts of force and power, efficient force application, and higher step frequencies.

**Key Words:** resisted sprinting, mechanical profiling, horizontal resistance training

## Introduction

The ability to improve sprint acceleration performance is a key component of physical preparation for many athletes. Resistance training exercises, specifically sled pulling, are often used to improve sprint acceleration performance (26,34). Resisted sled sprinting (RSS) provides a mechanical overload (i.e., requires the generation of high horizontal ground reaction forces (GRF) and increases impulse against the supporting ground) and has been identified as an effective training tool (1,9,13,17,20,26).

Recent research suggested that manipulating the load in sled training may affect different phases of the sprint acceleration and therefore different aspects of the athletes' mechanical capabilities (i.e., theoretical maximal horizontal power output [ $P_{max}$ ], theoretical maximal horizontal force [ $F_0$ ], theoretical maximal horizontal velocity [ $v_0$ ], slope of the linear force-velocity relationship [ $S_{Fv}$ ], maximal ratio of horizontal-to-resultant force [ $RF_{max}$ ], and rate of decrease in RF [ $D_{rf}$ ]) (3,20,25). For example, Morin et al. (20) reported an increase in the  $F_0$  and  $RF_{max}$ , with a negative trivial effect on  $v_0$ , after 16 training sessions with an RSS load of 80% of body mass (BM), compared with unresisted sprint training (URS). From a mechanical perspective, the adaptations to early or late acceleration phase after RSS, with heavier ( $>20\%$  of BM) or lighter ( $\leq 20\%$  of BM) loads, respectively, would be the result of the function of loads (26). This is because the early

acceleration phase depends on the generation of high horizontal forces at low running velocities and the late acceleration phase depends on high running velocities at low horizontal forces (3,20). During sprint acceleration, it is crucial to also consider the spatiotemporal parameters, such as step frequency (SF), step length (SL), flight time (FT), contact time (CT), and kinematic (joint and segment angles) because they are critical determinants of performance. Frequently, a decrease in SL, SF, and FT and an increase in CT during acute sprint towing trials (i.e., loaded vs. URS) (6,16,18,31) or long-term RSS training with low loads (14,35) has been reported.

In most cross-sectional studies, the load for sled pulling is prescribed as a percentage of BM or as the load necessary for a desired reduction in maximum sprint velocity ( $v_{dec}$ ). Some coaches and practitioners traditionally recommended performing RSS pulling with relatively light loads, which correspond to  $\leq 10\%$  of BM, or loads that reduce running velocity by  $\leq 10\%$  (14). However, because of differences in strength, gender, and training history across athletes, the  $v_{dec}$  approach has been suggested as a more appropriate way to prescribe the loads for each individual in connection to %BM (7). Previous literature has also mainly focused on the effects of long-term RSS with different but mostly light loads (2,35). Kawamori et al. (14) found that 8 weeks of RSS training with loads of 10 and  $30\% v_{dec}$  (corresponding to 13.1 and 43% of BM, respectively) significantly enhanced the 10-m sprint time, but only the  $30\% v_{dec}$  improved the 5-m sprinting performance. In a recent review, Fernández-Galván et al. (12) even reported that RSS training with loads above or below 20%

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of BM was not superior to URS. In addition, it has been proposed that loads corresponding to a 10%  $v_{\text{dec}}$  compared with URS, would not imply substantial changes in sprint mechanics (1).

It has been identified that RSS training with an external load associated with the apex of the velocity-power relationship ( $L_{50\%v_{\text{dec}}}$ ), which maximizes the horizontal power production (ranging from 69 to 91% of BM for recreational athletes and from 70 to 96% of BM for high-level sprinters), enhances sprinting performance (7,15). The  $L_{50\%v_{\text{dec}}}$  is defined as a load, which corresponds to a decrement of  $v_0$  of 50% ( $50\%v_{\text{dec}}$ ), based on the parabolic velocity-power relationship (7). Escobar Álvarez et al. (11) examined the effect of 8 weeks of sled pulling with loads corresponding to  $50\%v_{\text{dec}}$  on sprint acceleration performance, as well as on sprint mechanics. They found significant performance improvements in 5 and 20 m and in mechanical properties related to sprinting ( $F_0$ ,  $v_0$ ,  $P_{\text{max}}$ ,  $RF_{\text{max}}$ ). Moreover, Cahill et al. (3) found that 8 weeks of RSS training with loads of 25 and  $50\%v_{\text{dec}}$  significantly improved the  $P_{\text{max}}$  and 20-m sprinting performance. In addition, loads that cause a 75%  $v_{\text{dec}}$  enhanced the  $F_0$ ,  $P_{\text{max}}$ , and 20-m sprinting performance, compared with preintervention. Lahti et al. (15) observed greater  $P_{\text{max}}$  and faster 10-m sprint time, without spatiotemporal changes, during the 3 first steps of sprint acceleration and maximal velocity phases, following 9 weeks of RSS training with loads of 50 and  $60\%v_{\text{dec}}$ . Conversely, Cross et al. (8) found minor effects on sprinting performance and trivial benefits on  $P_{\text{max}}$ , after 12 sessions of RSS training intervention with loads representing 50 and  $10\%v_{\text{dec}}$  in sprint-trained athletes. Nevertheless, in sprinting performance and underlying mechanical properties, a between-groups comparison with heavy loads, lighter loads, or URS showed unclear or only small effects (8,20); in addition, there is lack of evidence regarding the effects of these heavy loads on the spatiotemporal characteristics throughout the sprint acceleration phase.

More research involving different magnitude of external loads is required to establish whether RSS training with heavy loads is more effective than a traditional RSS training with light loads or without loads (1). To the authors' knowledge, no study has been conducted to examine the effect of long-term RSS training with  $L_{50\%v_{\text{dec}}}$  in contrast to a traditional low (10% of BM) and URS on sprinting performance and mechanical and spatiotemporal parameters throughout the 30-m acceleration phase. Especially, the inclusion of the spatiotemporal parameters in the analysis of the results could reveal crucial information on the mechanisms behind potential changes in performance. Therefore, this study aimed to examine the effect of a 6-week RSS training program with 2 different loads, such as  $50\%v_{\text{dec}}$  and 10% of BM (10% BM), and URS, on sprint acceleration performance, sprint mechanics, and spatiotemporal characteristics. We hypothesized that the  $50\%v_{\text{dec}}$  training group would improve sprint acceleration performance and underlying mechanical variables, more than a traditional RSS training with a light load or without a load, through the increased anteroposterior power and force application during sprint acceleration.

## Methods

### Experimental Approach to the Problem

Subjects performed 12 sessions, over 6 weeks of resisted sled pulling and URS. In preintervention tests, we assessed each subject's F-v profile across 2 maximal 30-m sprints. Subjects were matched for 30-m best times and randomly divided into 3 training groups ( $50\%v_{\text{dec}}$ , 10% BM, URS). Pretraining and posttraining

testing of sprint acceleration performance and related mechanical and spatiotemporal parameters were conducted, respectively. Before the training sessions, subjects were familiarized with the sled loading and training protocol. All sessions were conducted on the same synthetic track surface, and all subjects were wearing their sprinting spikes.

### Subjects

Twenty-seven (17 males and 10 females) trained sprinters volunteered to participate in this study. Subjects were divided into  $50\%v_{\text{dec}}$  (age:  $18.3 \pm 0.7$  years; height:  $1.77 \pm 0.06$  m; BM:  $65.6 \pm 9.7$  kg; 30-m sprint time:  $4.64 \pm 0.23$  seconds), 10% BM (age:  $18.9 \pm 1.5$  years; height:  $1.75 \pm 0.09$  m; BM:  $64.2 \pm 7.4$  kg; 30-m sprint time:  $4.65 \pm 0.39$  seconds), and URS (age:  $20.3 \pm 3.6$  years; height:  $1.74 \pm 0.07$  m; BM:  $62.1 \pm 6.9$  kg; 30-m sprint time:  $4.68 \pm 0.35$  seconds) groups. The experimental procedures were approved by the National and Kapodistrian University of Athens Ethical Committee (approval protocol number: 1204/15-07-2020), in accordance with the Declaration of Helsinki, and all subjects gave their written informed consent to participate. Parental consent was obtained for those  $<18$  years of age. Subjects were free from lower extremity injury for at least 3 months before testing and had at least 1 year of experience in RSS training. Subjects were asked not to be involved in any other type of resistance training during the study.

### Procedures

One week before and 1 week after the first and last training sessions, subjects performed 2 maximal 30-m sprints from a 3-point crouching position. Three high-speed (300 Hz) panning cameras (Casio EX-F1, Tokyo, Japan) were used to continuously record the 30-m runway. The cameras were placed perpendicular to the running direction at the distances of 5, 15, and 25 m. The cameras had overlapping fields of view, and each camera recorded about 10 m of the sprint (6,32). Marking poles were placed at adjusted positions along the entire sprint distance to correctly capture the 5-m split times (27). In addition, for the extraction of SL, pairs of  $5 \times 5$  cm custom reference markers were set on either side of the running lanes, forming 1-m zones over the entire 30-m distance.

### Training Protocol

To determine the load-velocity (L-v) relationship, the subjects in the  $50\%v_{\text{dec}}$  group performed, before the first testing session, 1 unresisted and 3 resisted 30-m all-out sprints pulling a sled attached to their waists with a load corresponding to 25, 50, and 75% of BM, separated by 5 minutes of recovery (7). The maximal running velocity was obtained for URS and resisted sprint trials and was plotted against load to obtain an individualized L-v profile for each subject. The calculation of the  $L_{50\%v_{\text{dec}}}$  was performed using a validated Excel spreadsheet, based on the decrease of running speed through a linear L-v relationship (7,8,19). After a standardized sprint-specific warm-up, lasting approximately 30 minutes, all groups performed the same number of repetitions and covered the same distance, twice a week, during 6 consecutive weeks (12 sessions). During each training session, the groups performed 2 sets of five 20-m sprints, with 2 minutes of rest between sprints and 5 minutes of rest between the sets (20). The  $50\%v_{\text{dec}}$  group trained pulling a resisted sled attached to

their waist with a load corresponding to a 50% of reduction of individual  $v_0$  ( $43.1 \pm 9.0$  kg, ranging from 58 to 72% of BM). Accordingly, the 10% BM group trained with a load equal to 10% of BM ( $6.4 \pm 0.8$  kg), and the URS group trained with no resistance. Subjects initiated sprinting from a standing split stance with their lead foot placed behind the start line and were verbally encouraged to ensure that maximal effort was attained for each trial. Apart from training intervention, all subjects followed a similar weekly training programs but maximal sprinting efforts or sprint-specific and strength training was not allowed.

### Statistical Analyses

The split times per 5 m and spatiotemporal parameters were obtained from a video analysis software (Quintic Biomechanics v31; Consultancy Ltd., Birmingham, United Kingdom). The first propulsive movement of the back leg was considered as the starting point. The split times of 5, 10, 15, 20, 25, and 30 m were defined as the moment in which the subject's hip crossed the corresponding pole, and running speed was computed for each 5-m distance interval (27,30). The velocity-time data of the best pretraining and posttraining 30-m trials were used to determine the variables of the sprint mechanical profile:  $F_0$ ,  $v_0$ ,  $P_{\max}$ ,  $S_{Fv}$ ,  $RF_{\max}$ , and  $D_{rf}$  according to the method proposed by Samozino et al. (22,30). In addition, the moment of touchdown and toe-off of each step was defined as the frames at which the subject's foot made and lost contact with the ground, respectively. The CT was determined as the time interval between initial foot contact and toe-off and FT as the time interval from toe-off of the ipsilateral foot to touchdown of the contralateral foot. Step length for each step was defined by the locations at which the subject's toes contacted the ground during a stride. The distance from the toe to the nearest pairs of reference markers was determined by projecting the position of the subject's toe during the touchdown onto a line between 2 reference markers (4,10,23,32,36). Step frequency was calculated by dividing running velocity by SL. All spatiotemporal variables were presented as means per 5-m distance intervals. The accuracy of the data collection and of the computed spatiotemporal variables was examined previously and provided a very high intraclass correlation coefficient (0.999) (24).

### Statistical Analyses

All outcome measures are presented as mean  $\pm$  SD. Normality (Shapiro-Wilk test) and homogeneity of variance (Levene's test) were checked before analyses. A 3 (conditions: 50% $v_{\text{dec}}$ , 10% BM, and URS)  $\times$  2 (time: pre and post) mixed-model repeated measures analysis of variance was used to determine the within-group and between-group effects on running performance (time to 5, 10, 15, 20, 25, and 30 m), sprint mechanics ( $F_0$ ,  $v_0$ ,  $P_{\max}$ ,  $S_{Fv}$ ,  $RF_{\max}$ , and  $D_{rf}$ ) and spatiotemporal parameters (SL, SF, CT, and FT) at each interval distance and for each dependent variable. Bonferroni's post hoc comparisons were performed when applicable to examine interaction effects. Cohen's  $d$  effect size (ES) was used to assess the magnitude between pretraining to posttraining effects with values of <0.20, 0.20, 0.60, and 1.20 representing the qualitative thresholds for trivial, small, moderate, and large effects, respectively (5). The alpha level was set as  $p < 0.05$  with analysis performed in SPSS v28 (IBM Corp., Armonk, NY).

## Results

The results of the analysis of variance indicated a significant interaction effect (condition  $\times$  time) for the 5-m ( $F = 15.47$ ,  $p < 0.001$ ,  $\eta^2 = 0.56$ ), 10-m ( $F = 16.86$ ,  $p < 0.001$ ,  $\eta^2 = 0.58$ ), 15-m ( $F = 13.17$ ,  $p < 0.001$ ,  $\eta^2 = 0.52$ ), 20-m ( $F = 13.23$ ,  $p < 0.001$ ,  $\eta^2 = 0.52$ ), 25-m ( $F = 10.83$ ,  $p < 0.001$ ,  $\eta^2 = 0.47$ ), and 30-m ( $F = 8.27$ ,  $p = 0.002$ ,  $\eta^2 = 0.41$ ) sprint time. Bonferroni's post hoc interaction effect analysis revealed that only the 50% $v_{\text{dec}}$  group performed significantly faster sprint times at 5 m (mean difference [MD] =  $-0.072$  seconds,  $p < 0.001$ ), 10 m (MD =  $-0.084$  seconds,  $p < 0.001$ ), 15 m (MD =  $-0.097$  seconds,  $p < 0.001$ ), 20 m (MD =  $-0.113$  seconds,  $p < 0.001$ ), 25 m (MD =  $-0.127$  seconds,  $p < 0.001$ ) and 30 m (MD =  $-0.129$  seconds,  $p < 0.001$ ). No significant ( $p > 0.05$ ) interaction or main condition and time effects were observed in sprint acceleration performance for the 10% BM and URS groups (Table 1).

A main effect of time revealed a significant higher  $F_0$  ( $F = 5.39$ ,  $p = 0.029$ ,  $\eta^2 = 0.18$ ),  $P_{\max}$  ( $F = 11.17$ ,  $p = 0.003$ ,  $\eta^2 = 0.32$ ), and  $RF_{\max}$  ( $F = 8.41$ ,  $p = 0.008$ ,  $\eta^2 = 0.26$ ). Post hoc analysis indicated that after RSS training, only the 50% $v_{\text{dec}}$  group significantly improved the  $F_0$  (MD =  $0.51$  N $\cdot$ kg $^{-1}$ ,  $p = 0.035$ ),  $P_{\max}$  (MD =  $1.46$  W $\cdot$ kg $^{-1}$ ,  $p = 0.008$ ), and  $RF_{\max}$  (MD =  $0.17\%$ ,  $p = 0.019$ ) compared with pretraining (Figure 1). No significant interaction or main effects of time and condition were found for  $v_0$ ,  $S_{Fv}$ , and  $D_{rf}$  in all groups. However, a trend of increased  $v_0$  (ES = 0.64,  $p = 0.064$ ) was noted with the 50% $v_{\text{dec}}$  group post training (Table 2).

A significant interaction effect occurred in running velocity at 0–5 m ( $F = 14.73$ ,  $p < 0.001$ ,  $\eta^2 = 0.55$ ) and 15–20 m ( $F = 3.83$ ,  $p = 0.036$ ,  $\eta^2 = 0.24$ ) distance intervals. Post hoc tests indicated that post training, only the 50% $v_{\text{dec}}$  group showed significantly increased running velocity at 0–5 m (MD =  $0.18$  m $\cdot$ s $^{-1}$ ,  $p < 0.001$ ) and 15–20 m (MD =  $0.24$  m $\cdot$ s $^{-1}$ ,  $p < 0.001$ ). In addition, a main effect of time indicated a significant increase in running velocity at the following intervals: 5–10 m ( $F = 7.70$ ,  $p = 0.011$ ,  $\eta^2 = 0.24$ ), 10–15 m ( $F = 10.21$ ,  $p = 0.004$ ,  $\eta^2 = 0.30$ ), 20–25 m ( $F = 8.83$ ,  $p = 0.007$ ,  $\eta^2 = 0.27$ ), and 25–30 m ( $F = 4.27$ ,  $p = 0.049$ ,  $\eta^2 = 0.15$ ). Post hoc analysis revealed that only the 50% $v_{\text{dec}}$  group achieved significantly higher running velocity postintervention at 5–10 m (MD =  $0.14$  m $\cdot$ s $^{-1}$ ,  $p = 0.005$ ), 10–15 m (MD =  $0.17$  m $\cdot$ s $^{-1}$ ,  $p = 0.005$ ), 20–25 m (MD =  $0.14$  m $\cdot$ s $^{-1}$ ,  $p = 0.013$ ), and 25–30 m (MD =  $0.11$  m $\cdot$ s $^{-1}$ ,  $p = 0.039$ ) intervals, compared with pre-intervention trials. Moreover, an interaction effect was noticed in SF at 0–5 m distance interval ( $F = 4.88$ ,  $p = 0.017$ ,  $\eta^2 = 0.29$ ). Post hoc analysis showed that only the 50% $v_{\text{dec}}$  group achieved higher SF after RSS training (MD =  $0.172$  Hz,  $p < 0.001$ ). Furthermore, a main effect of time indicated a significant higher SF at 10–15 m ( $F = 9.34$ ,  $p = 0.005$ ,  $\eta^2 = 0.28$ ), 15–20 m ( $F = 7.22$ ,  $p = 0.013$ ,  $\eta^2 = 0.23$ ), and 20–25 m ( $F = 9.05$ ,  $p = 0.006$ ,  $\eta^2 = 0.27$ ) distance intervals. Post hoc tests revealed that post training, only the 50% $v_{\text{dec}}$  group significantly reached higher SF at 10–15 m (MD =  $0.116$  Hz,  $p = 0.002$ ), 15–20 m (MD =  $0.134$  Hz,  $p = 0.002$ ), and 20–25 m (MD =  $0.86$  Hz,  $p = 0.009$ ) intervals, compared with baseline. However, a trend of higher SF at 5–10 m and 25–30 m intervals (ES = 0.65 and 0.70, respectively) was observed in the 50% $v_{\text{dec}}$  group post intervention. In addition, an interaction effect was observed in FT at 5–10 m and 10–15 m intervals ( $F = 7.20$ ,  $p = 0.004$ ,  $\eta^2 = 0.38$  and  $F = 3.84$ ,  $p = 0.036$ ,  $\eta^2 = 0.24$ , respectively). Post hoc analysis revealed that only the 50% $v_{\text{dec}}$  group showed significantly reduced FT posttraining at 5–10 m (MD =  $-0.008$

**Table 1****Pre-post changes in sprinting performance after a 6-week training intervention displayed by group.\*†**

| Groups                    | Distance    | 5 m                  | 10 m                 | 15 m                 | 20 m                 | 25 m                 | 30 m                 |
|---------------------------|-------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Sprinting performance (s) |             |                      |                      |                      |                      |                      |                      |
| 50% $v_{dec}$             | Pre         | 1.46 ± 0.08          | 2.19 ± 0.10          | 2.85 ± 0.12          | 3.47 ± 0.16          | 4.06 ± 0.19          | 4.64 ± 0.23          |
|                           | Post        | 1.39 ± 0.07‡         | 2.11 ± 0.10‡         | 2.75 ± 0.13‡         | 3.35 ± 0.16‡         | 3.94 ± 0.20‡         | 4.51 ± 0.24§         |
|                           | % $\Delta$  | -4.8%                | -3.8%                | -3.4%                | -3.3%                | -3.1%                | -2.8%                |
|                           | ES (95% CI) | 1.81 (0.70 to 2.88)  | 2.34 (1.02 to 3.62)  | 2.38 (1.05 to 3.68)  | 2.39 (1.05 to 3.69)  | 2.21 (0.95 to 3.45)  | 1.99 (0.81 to 3.13)  |
| 10% BM                    | Pre         | 1.40 ± 0.09          | 2.16 ± 0.14          | 2.83 ± 0.21          | 3.45 ± 0.27          | 4.06 ± 0.33          | 4.65 ± 0.39          |
|                           | Post        | 1.39 ± 0.09          | 2.15 ± 0.14          | 2.81 ± 0.20          | 3.43 ± 0.26          | 4.03 ± 0.32          | 4.62 ± 0.38          |
|                           | % $\Delta$  | -0.8%                | -0.7%                | -0.7%                | -0.7%                | -0.7%                | -0.7%                |
|                           | ES (95% CI) | 0.47 (-0.23 to 1.15) | 0.58 (-0.15 to 1.28) | 0.80 (0.02 to 1.54)  | 0.74 (-0.02 to 1.46) | 0.76 (0 to 1.5)      | 0.61 (-0.07 to 1.39) |
| URS                       | Pre         | 1.42 ± 0.09          | 2.18 ± 0.13          | 2.84 ± 0.19          | 3.47 ± 0.24          | 4.08 ± 0.30          | 4.68 ± 0.35          |
|                           | Post        | 1.42 ± 0.08          | 2.17 ± 0.12          | 2.83 ± 0.17          | 3.45 ± 0.22          | 4.05 ± 0.26          | 4.65 ± 0.31          |
|                           | % $\Delta$  | -0.2%                | -0.5%                | -0.6%                | -0.7%                | -0.7%                | -0.7%                |
|                           | ES (95% CI) | 0.15 (-0.51 to 0.81) | 0.22 (-0.45 to 0.88) | 0.38 (-0.31 to 1.05) | 0.53 (-0.18 to 1.22) | 0.54 (-0.18 to 1.23) | 0.64 (0.10 to 1.35)  |

\*ES = effect size; % $\Delta$  = percent change; 95% CI = 95% confidence intervals;  $v_{dec}$  = decrement of the theoretical horizontal velocity; BM = body mass; URS = unresisted sprint.

†Values are mean ± SD.

‡ $p < 0.01$  significantly different from baseline.§ $p < 0.05$  significantly different from baseline.

seconds,  $p = 0.003$ ) and 10–15 m (MD = -0.003 seconds,  $p = 0.022$ ). Moreover, a main effect of time indicated a decrease in FT at 20–25 m ( $F = 5.52$ ,  $p = 0.027$ ,  $\eta^2 = 0.19$ ) and 25–30 m ( $F = 7.38$ ,  $p = 0.012$ ,  $\eta^2 = 0.24$ ) distance intervals. Post hoc tests revealed that postintervention, only the 50% $v_{dec}$  group showed significantly decreased FT at 20–25 m (MD = -0.006 seconds,  $p = 0.005$ ) and 25–30 m (MD = -0.007 seconds,  $p = 0.003$ ) intervals, compared with preintervention tests. No significant interaction or main effect was observed in FT at 0–5 m and 15–20 m intervals. In addition, there were no significant interaction effects or main condition and time effects in SL and CT. Sprint spatiotemporal variables for all groups are displayed in Table 3.

## Discussion

This study examined the effect of a twice weekly, 6-week RSS (50% $v_{dec}$  and 10% BM), and URS training programs on sprinting performance, sprint mechanics, and spatiotemporal parameters,

throughout the acceleration phase. The study results supported our hypothesis, indicating that RSS training with a load corresponding to 50%  $v_{dec}$  of the individual's  $v_0$  significantly improved sprint acceleration performance (5, 10, 15, 20, 25, and 30 m). The sprinting performance did not improve in the 10% BM and URS groups, indicating the potential advantage of training with heavier loads in developing higher running velocities throughout the sprint acceleration phase in trained sprinters. Our findings are in line with previous studies that have revealed similar improvements in sprinting performance after RSS training with a load associated with the apex of the velocity-power relationship, which maximizes the horizontal power production during sled pulling (3,11,15).

In addition, as per our hypothesis, the subjects assigned to the 50% $v_{dec}$  group significantly improved the  $F_0$ ,  $P_{max}$ , and the  $RF_{max}$ . The variables  $F_0$  and  $RF_{max}$  reflect the early acceleration phase in which the development and application of horizontal force is the highest and is applied during low running velocities. The greater  $F_0$  of the 50% $v_{dec}$  group corresponds to the initial

**Table 2****Pre-post changes in sprint mechanical profile following a 6-week training intervention displayed by group.\*†**

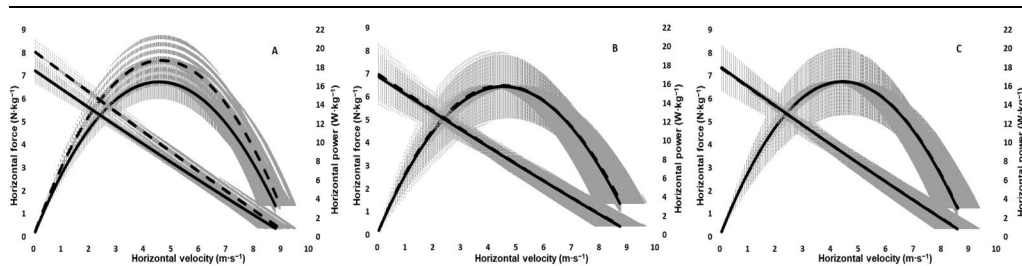
| Groups                    | Variable    | $F_0$ (N·kg <sup>-1</sup> ) | $v_0$ (m·s <sup>-1</sup> ) | $P_{max}$ (W·kg <sup>-1</sup> ) | $S_{Fv}$ (N·s·m <sup>-1</sup> ·kg <sup>-1</sup> ) | $RF_{max}$ (%)       | $D_{Ht}$ (%·s·m)    |
|---------------------------|-------------|-----------------------------|----------------------------|---------------------------------|---------------------------------------------------|----------------------|---------------------|
| Sprint mechanical profile |             |                             |                            |                                 |                                                   |                      |                     |
| 50% $v_{dec}$             | Pre         | 7.25 ± 0.46                 | 9.26 ± 0.76                | 16.79 ± 1.68                    | -0.79 ± 0.08                                      | 44.2 ± 1.7           | -7.3 ± 0.8          |
|                           | Post        | 7.76 ± 0.69‡                | 9.38 ± 0.71                | 18.26 ± 2.64‡                   | -0.83 ± 0.07                                      | 45.8 ± 2.6‡          | -7.7 ± 0.7          |
|                           | % $\Delta$  | 7.1%                        | 1.3%                       | 8.7%                            | 5.4%                                              | 3.7%                 | 4.7%                |
|                           | ES (95% CI) | 0.74 (-0.02 to 1.47)        | 0.64 (-0.10 to 1.35)       | 0.86 (0.07 to 1.62)             | 0.54 (0.18 to 1.23)                               | 0.93 (0.12 to 1.70)  | 0.44 (0.26 to 1.11) |
| 10% BM                    | Pre         | 6.87 ± 0.93                 | 9.23 ± 1.02                | 16.00 ± 3.36                    | -0.76 ± 0.07                                      | 42.9 ± 4.2           | -7.2 ± 0.4          |
|                           | Post        | 7.16 ± 1.23                 | 9.42 ± 0.85                | 16.85 ± 3.10                    | -0.76 ± 0.07                                      | 43.9 ± 3.4           | -7.4 ± 1.0          |
|                           | % $\Delta$  | 4.2%                        | 2.0%                       | 5.3%                            | 0%                                                | 2.2%                 | 2.4%                |
|                           | ES (95% CI) | 0.36 (-0.33 to 1.02)        | 0.40 (-0.29 to 1.07)       | 0.48 (-0.23 to 1.16)            | 0.09 (-0.57 to 0.74)                              | 0.42 (-0.28 to 1.09) | 0.33 (0.35 to 1.0)  |
| URS                       | Pre         | 7.15 ± 1.01                 | 9.05 ± 0.85                | 16.34 ± 3.74                    | -0.79 ± 0.07                                      | 43.5 ± 4.2           | -7.4 ± 0.6          |
|                           | Post        | 7.26 ± 0.97                 | 9.11 ± 0.80                | 16.68 ± 3.55                    | -0.80 ± 0.06                                      | 43.9 ± 3.9           | -7.4 ± 0.5          |
|                           | % $\Delta$  | 1.6%                        | 0.6%                       | 2.1%                            | 1.3%                                              | 1.0%                 | 0.6%                |
|                           | ES (95% CI) | 0.23 (-0.44 to 0.88)        | 0.24 (-0.44 to 0.89)       | 0.36 (-0.33 to 1.03)            | 0.16 (0.50 to 0.82)                               | 0.45 (-0.25 to 1.13) | 0.33 (0.35 to 1.0)  |

\* $F_0$  = theoretical maximal horizontal force;  $v_0$  = theoretical maximal horizontal velocity;  $P_{max}$  = theoretical maximal horizontal power;  $S_{Fv}$  = slope of the linear force-velocity relationship;  $RF_{max}$  = maximal ratio of horizontal-to-resultant force;  $D_{Ht}$  = rate of decrease in the ratio of horizontal force; ES = effect size; % $\Delta$  = percent change; 95% CI = 95% confidence intervals;  $v_{dec}$  = decrement of the theoretical horizontal velocity; BM = body mass; URS = unresisted sprint.

†Values are mean ± SD.

‡ $p < 0.05$  significantly different from baseline.





**Figure 1.** Preintervention (black line) and postintervention (dashed line) changes in the sprint mechanical profile following a 6-week resisted sled-pulling training with loads corresponding to a decrement of 50% of the theoretical horizontal velocity capability (A), 10% of BM (B), and URS (C). Data are average values with grey surrounding area representing SD. BM = body mass; URS = unresisted sprint training.

push of the subject onto the ground at sprint acceleration, which reflects a higher capability of lower limbs to generate horizontal force (21). In addition, the higher  $RF_{max}$  achieved by the 50% $v_{dec}$  group after the intervention suggests a more efficient pattern of force application. The latter corresponds to the ability to efficiently orient the anteroposterior force at the first steps of the acceleration phase in relation to the total force produced (21). Accordingly, the increase of  $P_{max}$  in the same group exhibits an increased ability to transfer higher amounts of horizontal force throughout the early and late acceleration phases posttraining (21). Our results confirm that heavy sled towing selectively improves  $F_0$ ,  $RF_{max}$  (20), and  $P_{max}$  (3,8,11,15,19). Conversely, no significant changes in  $v_0$  and  $D_{ff}$  were found postintervention in any of the groups. This indicates that the capability to produce horizontal force at high running velocities and the ability to maintain a net horizontal force production throughout the sprint acceleration phase did not differ after 6 weeks of RSS or URS training. However, the ES (0.64) along with the statistical significance level ( $p = 0.064$ ) indicate a trend toward increasing  $v_0$  in the 50% $v_{dec}$  group postintervention. It should also be mentioned that posttraining, the  $v_0$  increased in 7 of 9 subjects of the 50% $v_{dec}$  group. Moreover, the  $L_{50\%v_{dec}}$ , which maximizes the horizontal  $P_{max}$ , corresponds to half of the individual's  $F_0$  and  $v_0$ ; therefore, training with this load would be, in theory, beneficial in developing either  $F_0$  or  $v_0$ . Nevertheless, the steeper slope of the F-v relationship, which was displayed in the 50% $v_{dec}$  group after intervention (Figure 1), indicates that the greater part of the mechanical adaptations is expressed on the side of the horizontal force axis. The above may be explained by the high external resistance load of the 50% $v_{dec}$  group ( $65.3 \pm 5.3\%$  of BM), which is in agreement with Petrakos et al. (26) who concluded that heavy sled loads ( $>20\%$  of BM) improve the early acceleration phase. Furthermore, because of the orientation of the individuals' sprint mechanical profile (force-oriented or velocity-oriented profile), the adaptations of the RSS training with the  $L_{50\%v_{dec}}$  may be expressed differently, either by enhancing the development of horizontal force or by enhancing the velocity capabilities in each individual (7).

Finally, the mechanical properties of the sprint acceleration phase did not significantly change in the 10% BM and URS groups after the training intervention. This is in agreement with Petrakos et al. (26) who concluded that training with light loads and URS may not provide a sufficient stimulus for improving the sprint acceleration phase especially in already well-trained athletes. The latter could be derived from the fact that RSS training provides a sufficient stimulus for improving sprint acceleration performance because it causes mechanical adaptations, such as higher horizontal force generation capability and greater

mechanical effectiveness of force application, and not because it is specific to sprint patterns (14). A sufficient stimulus for improving the sprint acceleration phase could be provided because of the additional resistive stimulus at heavier loads during RSS towing in anteroposterior direction and/or the potential of subjects to increase the efficiency of force application.

The study results suggest that 6 weeks of RSS training with heavy load constitute an effective stimulus for increasing the SF during sprint acceleration, which might be caused by the improvement of the intermuscular and intramuscular coordination. According to Weyand et al. (33), faster running velocities are reached by applying higher GRF and not by repositioning the lower limbs more rapidly in the air. Our results show that there were no significant changes in SL and CT in the 3 training groups post intervention. By contrast, previous studies reported significant changes in SL (2) and CT (31) after RSS training with light loads. Furthermore, this study showed a decrease in FT only in the 50% $v_{dec}$  group. Theoretically, decreasing FT, while CT and SL remain constant, will cause increased running velocity as a result of an increased SF. The 50% $v_{dec}$  group demonstrated increases in SF as a result of decreases in FT. A decreased FT without the corresponding decrease in SL suggests differences in the orientation of the force vector during the late stance or propulsion phase. The above is supported by the increased percentage of horizontal forces in relation to the resultant. Kinematically, such changes could be explained by a possible increase of the forward lean of the overall body and trunk in the 50% $v_{dec}$  group. The possible changes in trunk lean after RSS training with heavy loads may balance the adaptations in SL and enhance the rate of sprint acceleration (35). Therefore, the higher mean SF achieved by the 50% $v_{dec}$  group after the intervention seems to be the reason behind performance differences during the early and late acceleration phases.

The higher SF could be explained by possible changes in the neural pathways and the subsequent increased rate of activation or by physiological changes that allow an increased firing rate of the cross-bridges cycle within the fast muscle fibers (29). Therefore, RSS training with heavy loads could possibly affect the maximal firing rate of tension development and the motor nerve conduction velocity of the agonist leg muscles, which contributes to a faster leg turnover (28,29). However, it is important to state that the adaptations could differ between SF-reliant and SL-reliant subjects; therefore, the individual's adaptation to the training stimuli may depend on different initial profiles (29).

The study has some limitations that must be addressed. First, in this study, we chose to control the total sprint training distance between the groups and thus keep 1 factor of the overall work

**Table 3****Pre-post changes in average values of spatiotemporal variables for 5 m distance intervals following a 6-week training intervention displayed by group.\*†**

| Groups                                | Distance    | 0–5 m                | 5–10 m               | 10–15 m              | 15–20 m              | 20–25 m              | 25–30 m              |
|---------------------------------------|-------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Running velocity (m·s <sup>-1</sup> ) |             |                      |                      |                      |                      |                      |                      |
| 50%v <sub>dec</sub>                   | Pre         | 3.44 ± 0.19          | 6.81 ± 0.29          | 7.64 ± 0.34          | 8.11 ± 0.45          | 8.44 ± 0.52          | 8.65 ± 0.57          |
|                                       | Post        | 3.61 ± 0.20‡         | 6.95 ± 0.37§         | 7.81 ± 0.42§         | 8.35 ± 0.50‡         | 8.58 ± 0.56§         | 8.75 ± 0.57§         |
|                                       | %Δ          | 5.1%                 | 2.0%                 | 2.2%                 | 2.9%                 | 1.7%                 | 1.2%                 |
|                                       | ES (95% CI) | 1.90 (0.76 to 3.01)  | 0.81 (0.03 to 1.56)  | 0.88 (0.09 to 1.64)  | 1.44 (0.47 to 2.37)  | 0.78 (0.01 to 1.52)  | 0.85 (0.06 to 1.61)  |
|                                       | ES (95% CI) | 0.51 (–0.20 to 1.20) | 0.22 (–0.45 to 0.87) | 0.36 (–0.33 to 1.02) | 0.38 (–0.31 to 1.05) | 0.34 (0.34 to 1.01)  | 0.05 (–0.61 to 0.70) |
| 10% BM                                | Pre         | 3.57 ± 0.20          | 6.64 ± 0.52          | 7.56 ± 0.67          | 8.05 ± 0.72          | 8.34 ± 0.78          | 8.55 ± 0.80          |
|                                       | Post        | 3.60 ± 0.23          | 6.66 ± 0.50          | 7.62 ± 0.65          | 8.12 ± 0.70          | 8.40 ± 0.75          | 8.56 ± 0.78          |
|                                       | %Δ          | 0.9%                 | 0.3%                 | 0.8%                 | 0.9%                 | 0.7%                 | 0.1%                 |
|                                       | ES (95% CI) | 0.51 (–0.20 to 1.20) | 0.22 (–0.45 to 0.87) | 0.36 (–0.33 to 1.02) | 0.38 (–0.31 to 1.05) | 0.34 (0.34 to 1.01)  | 0.05 (–0.61 to 0.70) |
|                                       | ES (95% CI) | 0.09 (–0.57 to 0.74) | 0.46 (–0.25 to 1.13) | 0.58 (–0.14 to 1.29) | 0.54 (–0.18 to 1.23) | 0.59 (–0.14 to 1.29) | 0.44 (–0.26 to 1.11) |
| URS                                   | Pre         | 3.53 ± 0.22          | 6.66 ± 0.51          | 7.54 ± 0.66          | 8.01 ± 0.68          | 8.26 ± 0.75          | 8.42 ± 0.78          |
|                                       | Post        | 3.53 ± 0.19          | 6.72 ± 0.46          | 7.61 ± 0.61          | 8.06 ± 0.61          | 8.34 ± 0.66          | 8.48 ± 0.70          |
|                                       | %Δ          | 0.1%                 | 0.8%                 | 1.0%                 | 0.7%                 | 0.9%                 | 0.7%                 |
|                                       | ES (95% CI) | 0.09 (–0.57 to 0.74) | 0.46 (–0.25 to 1.13) | 0.58 (–0.14 to 1.29) | 0.54 (–0.18 to 1.23) | 0.59 (–0.14 to 1.29) | 0.44 (–0.26 to 1.11) |
|                                       | ES (95% CI) | 0.09 (–0.57 to 0.74) | 0.46 (–0.25 to 1.13) | 0.58 (–0.14 to 1.29) | 0.54 (–0.18 to 1.23) | 0.59 (–0.14 to 1.29) | 0.44 (–0.26 to 1.11) |
| Step frequency (Hz)                   |             |                      |                      |                      |                      |                      |                      |
| 50%v <sub>dec</sub>                   | Pre         | 2.92 ± 0.38          | 4.24 ± 0.29          | 4.22 ± 0.21          | 4.22 ± 0.25          | 4.21 ± 0.26          | 4.16 ± 0.24          |
|                                       | Post        | 3.09 ± 0.30§         | 4.33 ± 0.34          | 4.34 ± 0.13§         | 4.35 ± 0.24§         | 4.30 ± 0.26§         | 4.23 ± 0.27          |
|                                       | %Δ          | 5.8%                 | 2.3%                 | 2.8%                 | 3.2%                 | 2.0%                 | 1.7%                 |
|                                       | ES (95% CI) | 1.21 (0.32 to 2.07)  | 0.65 (–0.09 to 1.36) | 0.97 (0.15 to 1.76)  | 1.15 (0.27 to 1.98)  | 0.98 (0.15 to 1.76)  | 0.70 (–0.05 to 1.42) |
|                                       | ES (95% CI) | 0.14 (–0.53 to 0.79) | 0.10 (–0.56 to 0.75) | 0.21 (–0.46 to 0.87) | 0.01 (–0.65 to 0.66) | 0.53 (–0.19 to 1.21) | 0.18 (–0.49 to 0.83) |
| 10% BM                                | Pre         | 3.15 ± 0.21          | 4.47 ± 0.35          | 4.42 ± 0.33          | 4.43 ± 0.27          | 4.35 ± 0.21          | 4.33 ± 0.18          |
|                                       | Post        | 3.17 ± 0.20          | 4.45 ± 0.27          | 4.43 ± 0.32          | 4.43 ± 0.26          | 4.40 ± 0.24          | 4.36 ± 0.18          |
|                                       | %Δ          | 0.6%                 | –0.4%                | 0.2%                 | 0%                   | 1.0%                 | 0.7%                 |
|                                       | ES (95% CI) | 0.14 (–0.53 to 0.79) | 0.10 (–0.56 to 0.75) | 0.21 (–0.46 to 0.87) | 0.01 (–0.65 to 0.66) | 0.53 (–0.19 to 1.21) | 0.18 (–0.49 to 0.83) |
|                                       | ES (95% CI) | 0.14 (–0.53 to 0.79) | 0.10 (–0.56 to 0.75) | 0.21 (–0.46 to 0.87) | 0.01 (–0.65 to 0.66) | 0.53 (–0.19 to 1.21) | 0.18 (–0.49 to 0.83) |
| URS                                   | Pre         | 2.90 ± 0.19          | 4.16 ± 0.16          | 4.12 ± 0.19          | 4.15 ± 0.14          | 4.12 ± 0.18          | 4.11 ± 0.18          |
|                                       | Post        | 2.94 ± 0.18          | 4.18 ± 0.16          | 4.15 ± 0.14          | 4.18 ± 0.15          | 4.15 ± 0.16          | 4.13 ± 0.17          |
|                                       | %Δ          | 1.4%                 | 0.6%                 | 0.7%                 | 1.1%                 | 0.6%                 | 0.5%                 |
|                                       | ES (95% CI) | 1.18 (0.29 to 2.03)  | 0.40 (–0.29 to 1.07) | 0.38 (–0.25 to 1.13) | 0.45 (–0.25 to 1.13) | 0.27 (–0.40 to 0.93) | 0.20 (–0.47 to 0.86) |
|                                       | ES (95% CI) | 1.18 (0.29 to 2.03)  | 0.40 (–0.29 to 1.07) | 0.38 (–0.25 to 1.13) | 0.45 (–0.25 to 1.13) | 0.27 (–0.40 to 0.93) | 0.20 (–0.47 to 0.86) |
| Step length (m)                       |             |                      |                      |                      |                      |                      |                      |
| 50%v <sub>dec</sub>                   | Pre         | 1.19 ± 0.11          | 1.61 ± 0.09          | 1.81 ± 0.07          | 1.92 ± 0.09          | 2.00 ± 0.07          | 2.08 ± 0.08          |
|                                       | Post        | 1.17 ± 0.08          | 1.61 ± 0.08          | 1.80 ± 0.08          | 1.92 ± 0.08          | 2.00 ± 0.06          | 2.07 ± 0.06          |
|                                       | %Δ          | –1.1%                | –0.3%                | –0.2%                | –0.4%                | –0.4%                | –0.4%                |
|                                       | ES (95% CI) | 0.19 (–0.47 to 0.85) | 0.11 (–0.55 to 0.76) | 0.11 (–0.55 to 0.76) | 0.11 (–0.55 to 0.76) | 0.41 (–0.29 to 1.08) | 0.22 (–0.45 to 0.87) |
|                                       | ES (95% CI) | 0.19 (–0.47 to 0.85) | 0.11 (–0.55 to 0.76) | 0.11 (–0.55 to 0.76) | 0.11 (–0.55 to 0.76) | 0.41 (–0.29 to 1.08) | 0.22 (–0.45 to 0.87) |
| 10% BM                                | Pre         | 1.14 ± 0.10          | 1.49 ± 0.14          | 1.72 ± 0.16          | 1.84 ± 0.18          | 1.93 ± 0.18          | 1.98 ± 0.20          |
|                                       | Post        | 1.14 ± 0.09          | 1.50 ± 0.13          | 1.72 ± 0.14          | 1.82 ± 0.16          | 1.91 ± 0.18          | 1.96 ± 0.18          |
|                                       | %Δ          | –0.8%                | –0.8%                | 0%                   | –1.0%                | –1.0%                | –0.8%                |
|                                       | ES (95% CI) | 0.24 (–0.43 to 0.90) | 0.24 (–0.43 to 0.89) | 0 (–0.65 to 0.65)    | 0.27 (–0.40 to 0.93) | 0.27 (–0.93 to 0.40) | 0.27 (–0.93 to 0.40) |
|                                       | ES (95% CI) | 0.24 (–0.43 to 0.90) | 0.24 (–0.43 to 0.89) | 0 (–0.65 to 0.65)    | 0.27 (–0.40 to 0.93) | 0.27 (–0.93 to 0.40) | 0.27 (–0.93 to 0.40) |
| URS                                   | Pre         | 1.21 ± 0.07          | 1.59 ± 0.08          | 1.81 ± 0.11          | 1.91 ± 0.12          | 1.98 ± 0.12          | 2.02 ± 0.14          |
|                                       | Post        | 1.20 ± 0.06          | 1.59 ± 0.09          | 1.80 ± 0.10          | 1.90 ± 0.12          | 1.97 ± 0.11          | 2.01 ± 0.12          |
|                                       | %Δ          | –1.1%                | –0.2%                | –0.6%                | –0.7%                | –0.8%                | –0.8%                |
|                                       | ES (95% CI) | 0.55 (–0.17 to 1.24) | 0.18 (–0.48 to 0.84) | 0.40 (–0.29 to 1.07) | 0.29 (–0.38 to 0.95) | 0.43 (–0.27 to 1.11) | 0.37 (–0.32 to 1.04) |
|                                       | ES (95% CI) | 0.55 (–0.17 to 1.24) | 0.18 (–0.48 to 0.84) | 0.40 (–0.29 to 1.07) | 0.29 (–0.38 to 0.95) | 0.43 (–0.27 to 1.11) | 0.37 (–0.32 to 1.04) |
| Contract time (s)                     |             |                      |                      |                      |                      |                      |                      |
| 50%v <sub>dec</sub>                   | Pre         | 0.173 ± 0.019        | 0.142 ± 0.013        | 0.130 ± 0.011        | 0.122 ± 0.009        | 0.121 ± 0.010        | 0.120 ± 0.010        |
|                                       | Post        | 0.175 ± 0.023        | 0.143 ± 0.014        | 0.130 ± 0.010        | 0.124 ± 0.010        | 0.122 ± 0.009        | 0.121 ± 0.011        |
|                                       | %Δ          | 0.9%                 | 0.4%                 | 0.6%                 | 1.4%                 | 0.8%                 | 1.0%                 |
|                                       | ES (95% CI) | 0.18 (–0.48 to 0.84) | 0.11 (–0.55 to 0.76) | 0.18 (–0.49 to 0.83) | 0.46 (–0.24 to 1.14) | 0.38 (–0.31 to 1.05) | 0.26 (–0.41 to 0.92) |
|                                       | ES (95% CI) | 0.18 (–0.48 to 0.84) | 0.11 (–0.55 to 0.76) | 0.18 (–0.49 to 0.83) | 0.46 (–0.24 to 1.14) | 0.38 (–0.31 to 1.05) | 0.26 (–0.41 to 0.92) |
| 10% BM                                | Pre         | 0.167 ± 0.017        | 0.136 ± 0.011        | 0.128 ± 0.007        | 0.120 ± 0.006        | 0.118 ± 0.006        | 0.117 ± 0.006        |
|                                       | Post        | 0.171 ± 0.013        | 0.134 ± 0.012        | 0.126 ± 0.006        | 0.120 ± 0.005        | 0.117 ± 0.006        | 0.115 ± 0.006        |
|                                       | %Δ          | 2.2%                 | –0.9%                | –1.5%                | –0.3%                | –1.2%                | –1.7%                |
|                                       | ES (95% CI) | 0.35 (–0.33 to 1.02) | 0.29 (–0.39 to 0.95) | 0.47 (–0.23 to 1.15) | 0.06 (–0.60 to 0.71) | 0.54 (–0.18 to 1.23) | 0.94 (–0.13 to 1.72) |
|                                       | ES (95% CI) | 0.35 (–0.33 to 1.02) | 0.29 (–0.39 to 0.95) | 0.47 (–0.23 to 1.15) | 0.06 (–0.60 to 0.71) | 0.54 (–0.18 to 1.23) | 0.94 (–0.13 to 1.72) |
| URS                                   | Pre         | 0.185 ± 0.012        | 0.148 ± 0.008        | 0.135 ± 0.008        | 0.128 ± 0.009        | 0.125 ± 0.009        | 0.124 ± 0.009        |
|                                       | Post        | 0.183 ± 0.010        | 0.146 ± 0.007        | 0.132 ± 0.007        | 0.126 ± 0.009        | 0.122 ± 0.009        | 0.120 ± 0.008        |
|                                       | %Δ          | –1.1%                | –1.5%                | –2.2%                | –0.9%                | –2.2%                | –2.5%                |
|                                       | ES (95% CI) | 0.16 (–0.51 to 0.81) | 0.29 (0.39 to 0.95)  | 0.64 (–0.10 to 1.34) | 0.24 (–0.43 to 0.89) | 0.57 (–0.15 to 1.27) | 0.72 (–0.04 to 1.44) |
|                                       | ES (95% CI) | 0.16 (–0.51 to 0.81) | 0.29 (0.39 to 0.95)  | 0.64 (–0.10 to 1.34) | 0.24 (–0.43 to 0.89) | 0.57 (–0.15 to 1.27) | 0.72 (–0.04 to 1.44) |
| Flight time (s)                       |             |                      |                      |                      |                      |                      |                      |
| 50%v <sub>dec</sub>                   | Pre         | 0.081 ± 0.012        | 0.100 ± 0.010        | 0.109 ± 0.006        | 0.115 ± 0.010        | 0.120 ± 0.010        | 0.125 ± 0.009        |
|                                       | Post        | 0.074 ± 0.013        | 0.093 ± 0.009§       | 0.105 ± 0.006§       | 0.110 ± 0.007        | 0.114 ± 0.007§       | 0.118 ± 0.007§       |
|                                       | %Δ          | –9.6%                | –7.7%                | –3.3%                | –4.2%                | –4.9%                | –5.8%                |
|                                       | ES (95% CI) | 0.51 (–0.20 to 1.19) | 1.01 (0.18 to 1.88)  | 0.60 (–0.13 to 1.30) | 0.56 (–0.16 to 1.26) | 0.94 (0.13 to 1.72)  | 1.25 (0.34 to 2.12)  |
|                                       | ES (95% CI) | 0.51 (–0.20 to 1.19) | 1.01 (0.18 to 1.88)  | 0.60 (–0.13 to 1.30) | 0.56 (–0.16 to 1.26) | 0.94 (0.13 to 1.72)  | 1.25 (0.34 to 2.12)  |
| 10% BM                                | Pre         | 0.072 ± 0.015        | 0.089 ± 0.014        | 0.100 ± 0.011        | 0.107 ± 0.012        | 0.112 ± 0.012        | 0.117 ± 0.008        |
|                                       | Post        | 0.064 ± 0.016        | 0.091 ± 0.014        | 0.102 ± 0.009        | 0.106 ± 0.009        | 0.109 ± 0.008        | 0.114 ± 0.008        |
|                                       | %Δ          | –10.4%               | 2.0%                 | 1.8%                 | –0.9%                | –2.5%                | –2.7%                |
|                                       | ES (95% CI) | 0.72 (–0.04 to 1.44) | 0.25 (–0.42 to 0.91) | 0.71 (–0.05 to 1.43) | 0.14 (–0.52 to 0.80) | 0.50 (–0.21 to 1.18) | 0.35 (–0.33 to 1.02) |
|                                       | ES (95% CI) | 0.72 (–0.04 to 1.44) | 0.25 (–0.42 to 0.91) | 0.71 (–0.05 to 1.43) | 0.14 (–0.52 to 0.80) | 0.50 (–0.21 to 1.18) | 0.35 (–0.33 to 1.02) |

**Table 3****Pre-post changes in average values of spatiotemporal variables for 5 m distance intervals following a 6-week training intervention displayed by group.\*† (Continued)**

| Groups | Distance    | 0–5 m                | 5–10 m               | 10–15 m              | 15–20 m              | 20–25 m              | 25–30 m              |
|--------|-------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| URS    | Pre         | 0.075 ± 0.014        | 0.094 ± 0.007        | 0.107 ± 0.006        | 0.111 ± 0.005        | 0.115 ± 0.006        | 0.119 ± 0.005        |
|        | Post        | 0.080 ± 0.007        | 0.098 ± 0.008        | 0.107 ± 0.006        | 0.112 ± 0.006        | 0.116 ± 0.005        | 0.119 ± 0.005        |
|        | %Δ          | 6.7%                 | 4.2%                 | −0.7%                | 1.0%                 | 0.6%                 | 0.1%                 |
|        | ES (95% CI) | 0.41 (−0.28 to 1.08) | 0.62 (−0.11 to 1.32) | 0.17 (−0.50 to 0.82) | 0.36 (−0.33 to 1.03) | 0.17 (−0.49 to 0.83) | 0.00 (−0.65 to 0.65) |

\*ES = effect size; %Δ = percent change; 95% CI = 95% confidence intervals;  $v_{dec}$  = decrement of the theoretical horizontal velocity; BM = body mass; URS = unresisted sprint.

†Values are mean ± SD.

‡ $p < 0.01$  significantly different from baseline.\$ $p < 0.05$  significantly different from baseline.

produced constant. However, the total time spent during sprinting (i.e., the time under tension) is greater in an RSS training with heavy load, compared with light load or without load. Consequently, the forces experienced by the subjects between different groups changed not only in magnitude but also in duration, which could have effects on the neuromuscular adaptations after the training. Second, it is important to acknowledge that prescribing a load based on %BM does not consider possible variations in individual strength among the subjects and does not account for the contribution of surface friction to the overall force required to accelerate the sled during RSS pulling. Although in the scope of our study (light vs. heavier loads), this does not affect our results, future studies should consider adopting a  $v_{dec}$  approach when appropriate.

In conclusion, the findings of this study confirm our hypothesis that 12 sessions of RSS training with loads corresponding to a decrement of 50% of the theoretical horizontal velocity capability enhance sprinting performance by improving the generation and application of horizontal force and power and by achieving higher step frequencies during the early and late phases of sprint acceleration.

### Practical Applications

Findings of this study suggest that 12 sessions of RSS training with loads corresponding to a 50% decrement of the  $v_0$  lead to a significant increase in sprint acceleration performance. The improvements in sprint acceleration performance at all split distances (ranging from 2.8 to 4.8%) were reflected through changes in the associated horizontal force, velocity, and power properties, as well as in SF. This indicates that training with a load associated with the apex of the velocity-power relationship may be an effective loading stimulus to enhance horizontal force and power production capacities, improve the efficiency of horizontal force application, and achieve faster leg turnover during sprint acceleration in sprinters. Our results also suggest that traditional sled (10% of BM) and URS may not provide a sufficient stimulus for improving the sprint acceleration phase, especially in trained sprinters with experience in RSS training. Practitioners may consider RSS training with heavier loads as a specialized form of resistance training, which provides adequate stimulus to foster beneficial adaptations in sprint acceleration performance.

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